NeuroLens (Clinical Assistant): An Interactive Tool for Efficient EEG Annotation

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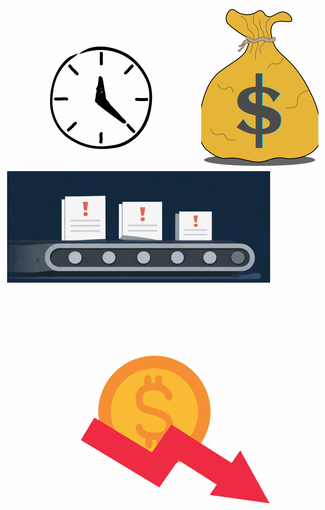
Project Description

Problem Statement

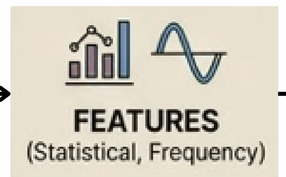
EEG-based seizure detection is limited by the scarcity of labeled data, as annotation requires clinical expert and is time-consuming and costly. This creates a bottleneck in building scalable and accurate machine learning models.

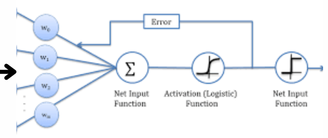
Solution (Active Learning)

Active learning addresses this by selecting only the most uncertain or informative samples for annotation, allowing the model to learn effectively with fewer labels while minimizing expert effort and labeling cost.



Dataset & Model



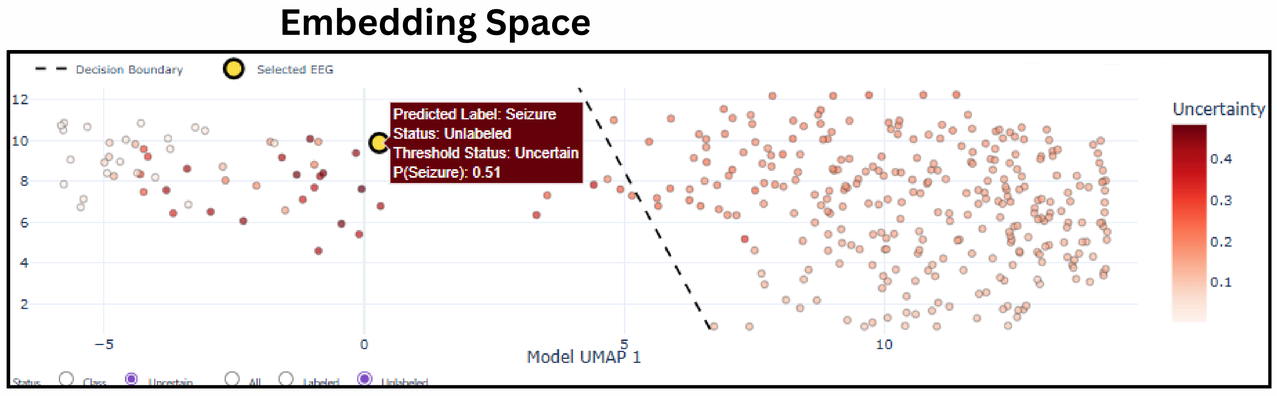


Logistic Regression
(Seizure vs Non-Seizure)

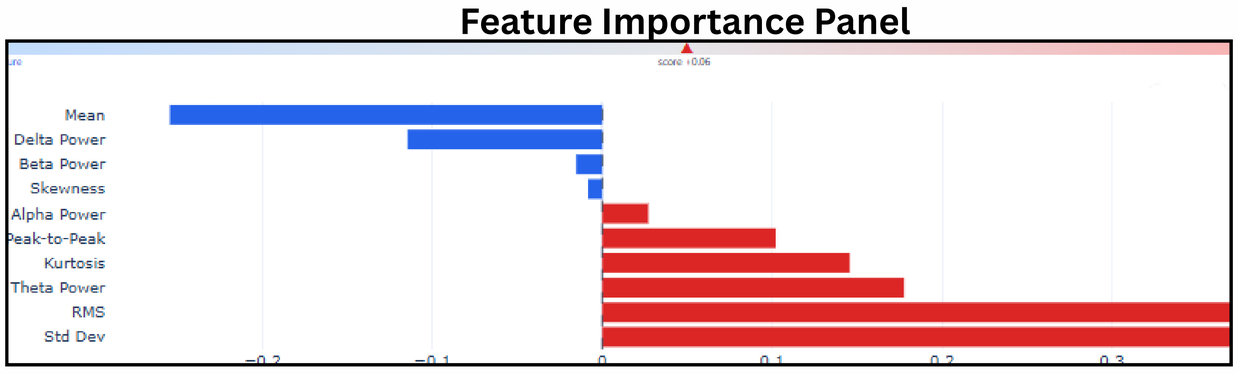
Workflow

(Start Criteria: Train on 40 labeled samples, Stop When: Model confident on k uncertain samples + No class flips)

Embedding Space



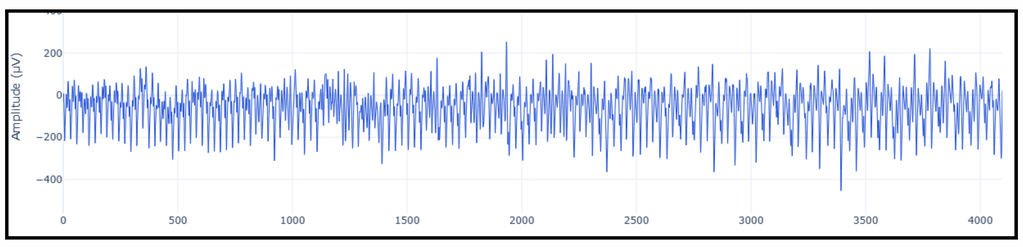
Feature Importance Panel



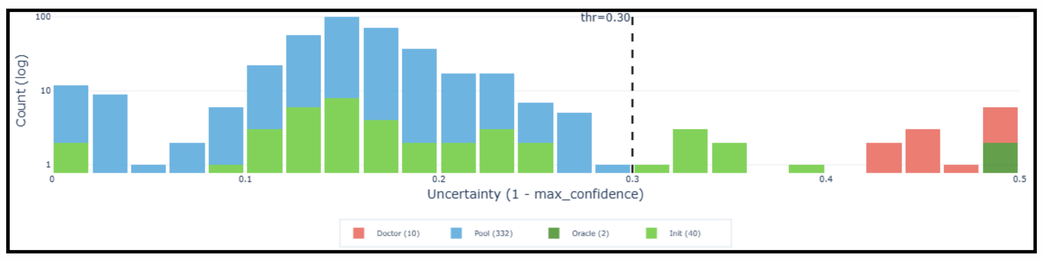
Select uncertain Samples → **Annotate (Expert)** → **Train Model** → **Model Updates**

Active Learning Loop

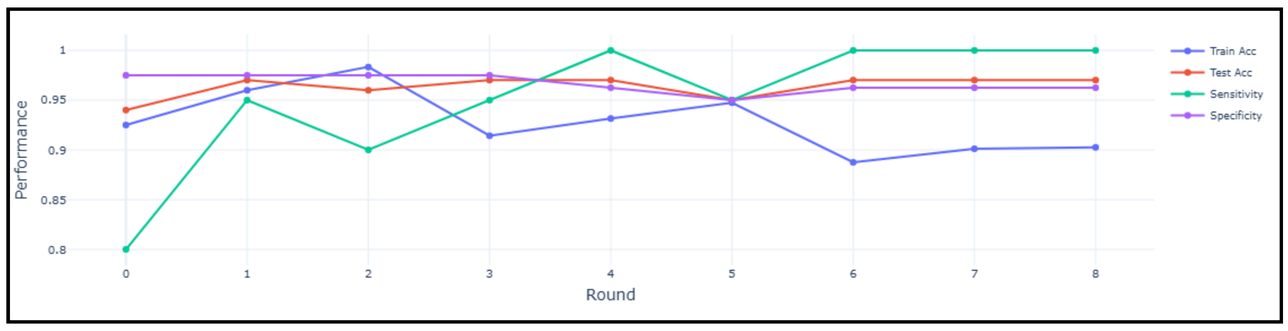
EEG Annotation Panel



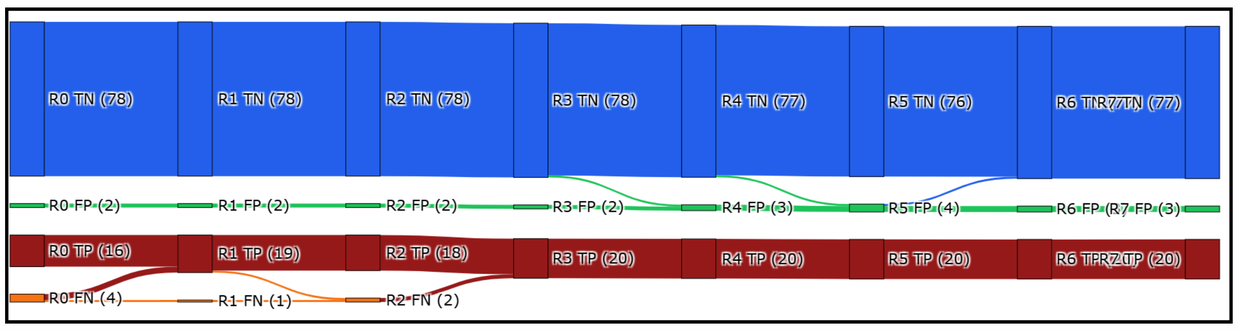
Uncertainty Histogram



Learning Curve

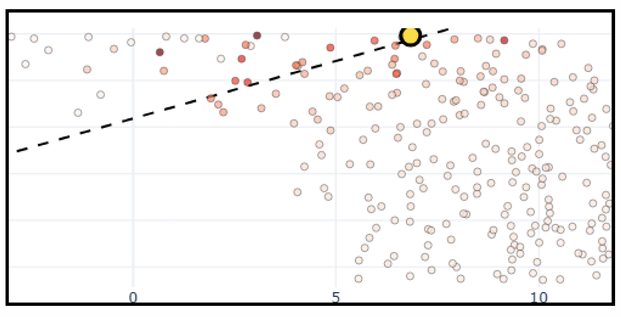


Prediction Flow Across Rounds

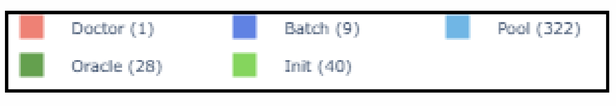


Key Findings

- **Test accuracy bypasses Train accuracy .**
- **Sensitivity bypasses Specificity.**
- Targeted **uncertainty** near **decision boundary**.

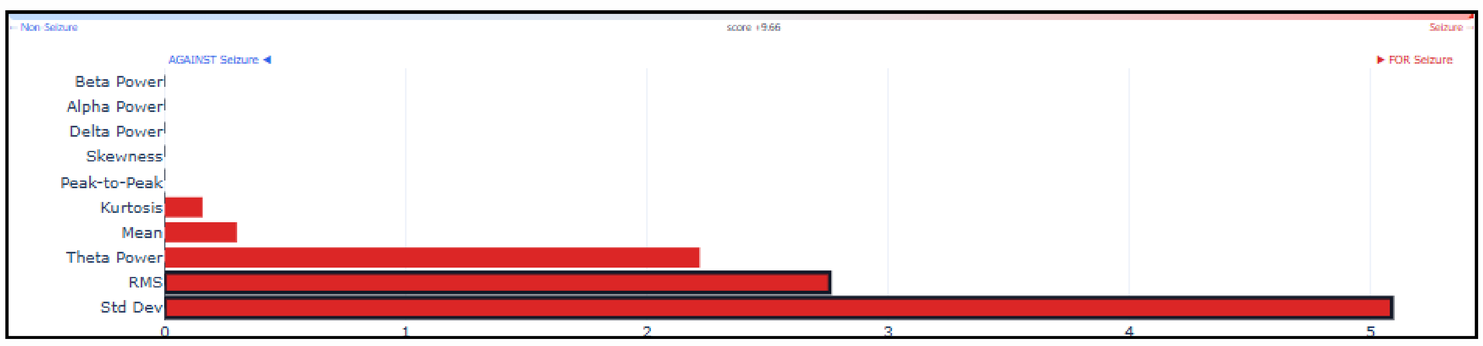


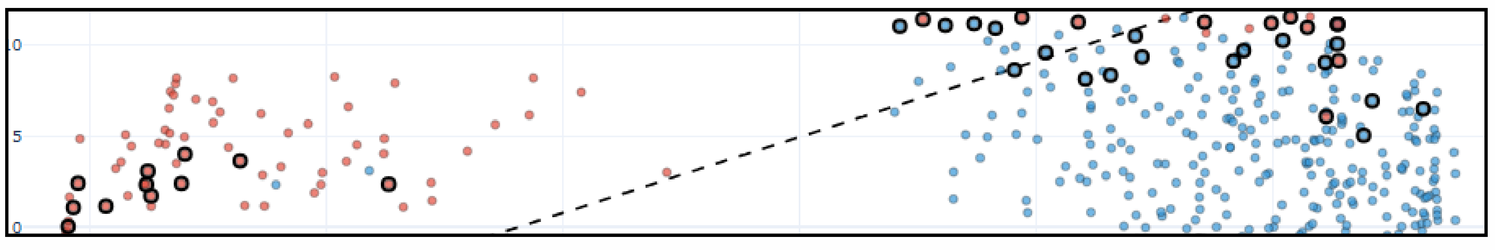
- As model learns **requirement of expert decreases.**

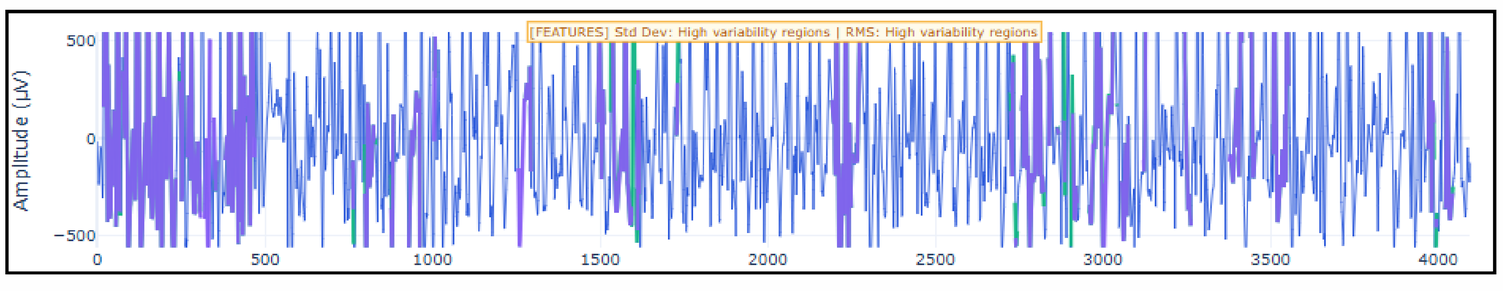
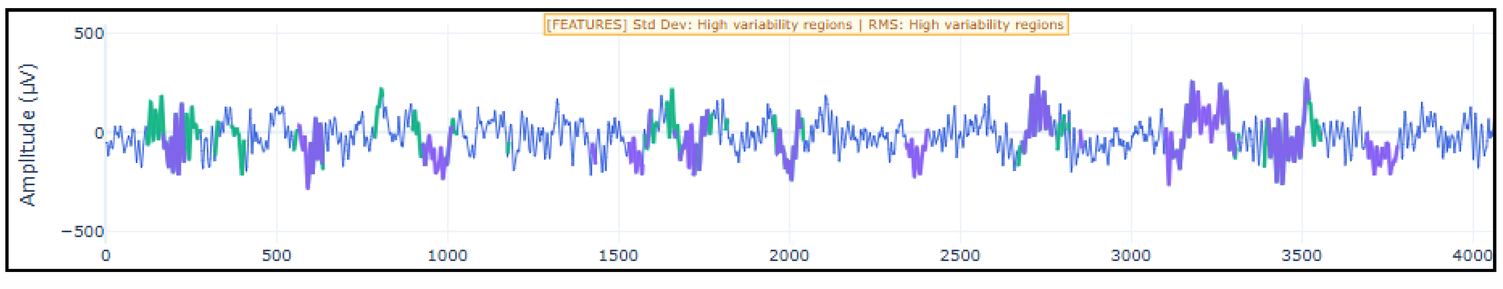


- Hybrid labeling (**targeted+top-k uncertain samples**) leads to faster and better convergence.
- **15-20%** of labeled data required only.

- **Findings match expert research and insights.**
- Overall major feature contributors**: RMS, Kurtosis, Theta Power, Mean, Std Dev**
- Majorly **seizure samples** require expert annotation.







- **Seizures are** associated with **higher amplitudes** and significantly **greater variability.**